

Course 4461

4 Credits: 4

Program Core

Source-grounded

# Strength of Materials (FS)

- To understand basic concepts of stress and strain in materials.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 4461 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 4461.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item                 | Official information                  |
|----------------------|---------------------------------------|
| Programme            | Diploma in Fire Technology and Safety |
| Course code          | 4461                                  |
| Course title         | Strength of Materials (FS)            |
| Semester             | 4 Credits: 4                          |
| Course category      | Program Core                          |
| Periods per week     | As specified in the official PDF      |
| Periods per semester | 60                                    |

| Item         | Official information                                 |
|--------------|------------------------------------------------------|
| Credits      | 4                                                    |
| Prerequisite | See the official syllabus PDF                        |
| Assessment   | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

16 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- To understand basic concepts of stress and strain in materials.
- To learn how to calculate shear force and bending moments in beams.
- To explore how beams bend and twist under load.
- To learn about torsion in shafts, spring deflection, and the stresses in thin cylinders.

#### Course outcomes

| CO  | Official outcome                                                          | Study level           |
|-----|---------------------------------------------------------------------------|-----------------------|
| CO1 | Explain and apply the concepts specified in the official course syllabus. | See official syllabus |

#### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3 2                    |
| CO2 | CO2 3 2                    |
| CO3 | CO3 3 3                    |
| CO4 | CO4 3 3                    |

### COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

#### Module coverage

| Module   | Official heading                                                          | Topic entries | Hours        |
|----------|---------------------------------------------------------------------------|---------------|--------------|
| Module 1 | Understand basic stress and strain and calculate changes in dimensions of | 5             | As specified |
| Module 2 | cases.                                                                    | 3             | As specified |
| Module 3 | Understand the bending of beams, and solve problems related to deflection | 5             | As specified |
| Module 4 | solve related problems.                                                   | 3             | As specified |

#### MODULE 1

## Understand basic stress and strain and calculate changes in dimensions of

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Understand basic stress and strain and calculate changes in dimensions of
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

## 2 Understanding and strain.

2 Understanding and strain. is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding and strain. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding and strain. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding and strain. means in the official course context             |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- 2 Understanding and strain.   
 definition/meaning . steps, application . Technical terms English-   
 .

## Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and

Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understand stress-strain diagrams for common 2 understanding materials. calculate stress and strain for uniform and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-  $\sigma$  Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and  $\epsilon$  definition/meaning  $\sigma$ .  $\epsilon$   $\sigma$   $\epsilon$ , steps, application  $\sigma$   $\epsilon$   $\sigma$ . Technical terms English-  $\sigma$   $\epsilon$   $\sigma$ .

## 2 Understanding composite sections. Understand thermal stresses in materials..

2 Understanding composite sections. Understand thermal stresses in materials.. is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding composite sections. Understand thermal stresses in materials.. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding composite sections. understand thermal stresses in materials.. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding composite sections. Understand thermal stresses in materials.. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-2 Understanding composite sections.

Understand thermal stresses in materials.. താപസമ്മർദ്ദം definition/meaning താപസമ്മർദ്ദം. താപസമ്മർദ്ദം താപസമ്മർദ്ദം, steps, application താപസമ്മർദ്ദം താപസമ്മർദ്ദം. Technical terms English-താപസമ്മർദ്ദം താപസമ്മർദ്ദം.

## 2 Understanding Solve problems on stress and strain.

2 Understanding Solve problems on stress and strain. is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Solve problems on stress and strain. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding solve problems on stress and strain. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding Solve problems on stress and strain. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-2 Understanding Solve problems on stress and strain. പഠിക്കേണ്ടതെല്ലാം പഠിച്ച് definition/meaning പഠിക്കേണ്ടതെല്ലാം. പഠിക്കേണ്ട പഠിക്കേണ്ട പഠിക്കേണ്ട, steps, application പഠിക്കേണ്ട പഠിക്കേണ്ടതെല്ലാം പഠിക്കേണ്ട. Technical terms English-പഠിക്കേണ്ട പഠിക്കേണ്ടതെല്ലാം.

## **5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI). Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple**

5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI). Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple is an official topic in Module 1 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI). Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, 5 applying types of stress and strain: tensile, compressive, and shear. stress-strain diagrams for materials like mild steel (ms) and cast iron (ci).thermal stress in materials calculate shear force and bending moments, and draw diagrams for simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI). Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- 5 Applying Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI). Thermal stress in materials Calculate shear force and bending moments, and draw diagrams for simple   
 definition/meaning   
 .   
 , steps, application   
 . Technical terms English-   
 .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Understand basic stress and strain and calculate changes in dimensions of materials.

|                                                    |                                              |   |
|----------------------------------------------------|----------------------------------------------|---|
| Learn types of forces and basic concepts of stress |                                              |   |
| M1.01                                              |                                              | 2 |
| Understanding                                      | and strain.                                  |   |
| M1.02                                              | Understand stress-strain diagrams for common | 2 |
| Understanding                                      | materials.                                   |   |
| Calculate stress and strain for uniform and        |                                              |   |
| M1.03                                              |                                              | 2 |
| Understanding                                      | composite sections.                          |   |
| Understand thermal stresses in materials..         |                                              |   |
| M1.04                                              |                                              | 2 |
| Understanding                                      |                                              |   |
| Solve problems on stress and strain.               |                                              |   |
| M1.05                                              |                                              | 5 |
| Applying                                           |                                              |   |

Contents:

Types of stress and strain: tensile, compressive, and shear.  
Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI). Thermal

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

### cases.

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.



**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: cases.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment

Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment is an official topic in Module 2 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understand different types of beams and loads. 3 understanding learn shear force and bending moment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- $\square$  Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment  $\square\square\square\square\square\square\square\square\square\square$   $\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square\square$   $\square\square\square\square\square$   $\square\square\square\square\square\square$ , steps, application  $\square\square\square\square\square$   $\square\square\square\square\square\square\square\square$   $\square\square\square\square\square$ . Technical terms English- $\square$   $\square\square\square\square$   $\square\square\square\square\square\square\square\square\square$ .

## 2 Understanding definitions Calculate shear force and bending moments for

2 Understanding definitions Calculate shear force and bending moments for is an official topic in Module 2 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding definitions Calculate shear force and bending moments for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding definitions calculate shear force and bending moments for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding definitions Calculate shear force and bending moments for means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2- 2 Understanding definitions Calculate shear force and bending moments for 2020 definition/meaning 2020. 2020 2020 2020, steps, application 2020 2020 2020. Technical terms English- 2020 2020.

**10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams.**

10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. is an official topic in Module 2 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

**Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

**Study points**

- Define or identify 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

**Engineering or professional relevance**

In Polytechnic practice, 10 applying various beam cases. types of beams: cantilever, simply supported, overhanging, etc. types of loads: point load, uniformly distributed load (udl). drawing shear force and bending moment diagrams. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect              | What to prepare                                                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams. **പ്രധാനപ്പെട്ട പദങ്ങൾ** definition/meaning **പ്രധാനപ്പെട്ട പദങ്ങൾ**. **പ്രധാനപ്പെട്ട പദങ്ങൾ** **പ്രധാനപ്പെട്ട പദങ്ങൾ**, steps, application **പ്രധാനപ്പെട്ട പദങ്ങൾ** **പ്രധാനപ്പെട്ട പദങ്ങൾ**. **Technical terms English-**പ്രധാനപ്പെട്ട പദങ്ങൾ****.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

cases.

|               |                                                |    |
|---------------|------------------------------------------------|----|
| M2.01         | Understand different types of beams and loads. | 3  |
| Understanding |                                                |    |
|               | Learn shear force and bending moment           |    |
| M2.02         |                                                | 2  |
| Understanding | definitions                                    |    |
|               | Calculate shear force and bending moments for  |    |
| M2.03         |                                                | 10 |
| Applying      | various beam cases.                            |    |
|               | Series Test 1                                  | 1  |

Contents:

Types of beams: cantilever, simply supported, overhanging, etc.

Types of loads: point load, uniformly distributed load (UDL).

Drawing shear force and bending moment diagrams.

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## Understand the bending of beams, and solve problems related to deflection

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes



## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Understand the bending of beams, and solve problems related to deflection
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding Understand the bending equation and apply it to

2 Understanding Understand the bending equation and apply it to is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Understand the bending equation and apply it to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding understand the bending equation and apply it to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding Understand the bending equation and apply it to means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3- 2 Understanding Understand the bending equation and apply it to  $\frac{1}{R} = \frac{Y}{\rho}$  definition/meaning  $\frac{1}{R} = \frac{Y}{\rho}$ .  $\frac{1}{R} = \frac{Y}{\rho}$   $\frac{1}{R} = \frac{Y}{\rho}$ , steps, application  $\frac{1}{R} = \frac{Y}{\rho}$   $\frac{1}{R} = \frac{Y}{\rho}$ . Technical terms English-  $\frac{1}{R} = \frac{Y}{\rho}$   $\frac{1}{R} = \frac{Y}{\rho}$ .

### 3 Understanding simple problems.

3 Understanding simple problems. is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding simple problems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding simple problems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding simple problems. means in the official course context        |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-3 Understanding simple problems.

പ്രശ്നങ്ങൾക്ക് definition/meaning നൽകുക. പ്രശ്നങ്ങൾക്ക് steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

## Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard

Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, calculate bending stress, modulus of section, and 6 applying moment of resistance. understand beam deflection and apply standard should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- $\sigma = \frac{M}{I} y$  Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard  $\delta = \frac{WL^3}{48EI}$  definition/meaning  $\delta = \frac{WL^3}{48EI}$ .  $\delta = \frac{WL^3}{48EI}$   $\delta = \frac{WL^3}{48EI}$ , steps, application  $\delta = \frac{WL^3}{48EI}$   $\delta = \frac{WL^3}{48EI}$ . Technical terms English- $\delta = \frac{WL^3}{48EI}$   $\delta = \frac{WL^3}{48EI}$ .

## 2 Understanding formulae. Learn about columns and struts, and calculate

2 Understanding formulae. Learn about columns and struts, and calculate is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding formulae. Learn about columns and struts, and calculate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding formulae. learn about columns and struts, and calculate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding formulae. Learn about columns and struts, and calculate means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related



but different term.

**Malayalam support:** Module 3-2 Understanding formulae. Learn about columns and struts, and calculate  $\frac{P}{A}$  definition/meaning  $\frac{P}{A}$ .  $\frac{P}{A}$   $\frac{P}{A}$   $\frac{P}{A}$ , steps, application  $\frac{P}{A}$   $\frac{P}{A}$ . Technical terms English-  $\frac{P}{A}$   $\frac{P}{A}$ .

### **3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler's formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and**

3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler's formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and is an official topic in Module 3 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### **Study points**

- Define or identify 3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler's formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### **Engineering or professional relevance**

In Polytechnic practice, 3 applying buckling load. bending stress and moment of resistance. deflection of beams under point load and udl.columns: euler's formula for buckling learn the behavior of shafts, springs, and thin cylinders under load, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| Aspect              | What to prepare                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler's formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 3 Applying buckling load. Bending stress and moment of resistance. Deflection of beams under point load and UDL. Columns: Euler's formula for buckling Learn the behavior of shafts, springs, and thin cylinders under load, and definition/meaning. Steps, application. Technical terms English- Malayalam.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Understand the bending of beams, and solve problems related to deflection and columns.

Learn the basic theory of bending.

|               |   |
|---------------|---|
| M3.01         | 2 |
| Understanding |   |

Understand the bending equation and apply it to

|               |   |
|---------------|---|
| M3.02         | 3 |
| Understanding |   |

simple problems.

|          |   |
|----------|---|
| M3.03    | 6 |
| Applying |   |

Calculate bending stress, modulus of section, and moment of resistance.

Understand beam deflection and apply standard

|               |   |
|---------------|---|
| M3.04         | 2 |
| Understanding |   |

formulae.

Learn about columns and struts, and calculate

|          |   |
|----------|---|
| M3.05    | 3 |
| Applying |   |

buckling load.

Contents:

Bending stress and moment of resistance.

Deflection of beams under point load and UDL. Columns: Euler's formula for

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 4

## solve related problems.

This module is presented from the official syllabus scope for Strength of Materials (FS). Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: solve related problems.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## 5 Understanding of inertia (M.I.). Understand how springs work and calculate their

5 Understanding of inertia (M.I.). Understand how springs work and calculate their is an official topic in Module 4 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding of inertia (M.I.). Understand how springs work and calculate their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding of inertia (m.i.). understand how springs work and calculate their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding of inertia (M.I.). Understand how springs work and calculate their means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-5 Understanding of inertia (M.I.). Understand how springs work and calculate their  $\Delta x$  definition/meaning  $\Delta x$ .  $\Delta x$   $\Delta x$   $\Delta x$ , steps, application  $\Delta x$   $\Delta x$ . Technical terms English- $\Delta x$   $\Delta x$   $\Delta x$ .

## 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and

5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and is an official topic in Module 4 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding deflection and stiffness. understand stresses in thin cylindrical shells and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and  $\sigma = \frac{p r}{t}$  definition/meaning  $\sigma$   $p$   $r$   $t$ .  $\sigma$   $p$   $r$   $t$ , steps, application  $\sigma$   $p$   $r$   $t$ . Technical terms English- $\sigma$   $p$   $r$   $t$ .

**4 Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press**

4 Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press is an official topic in Module 4 of Strength of Materials (FS). Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 4 Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 4 understanding calculate their thickness. seriestest-ii 1 torsion in shafts: calculation of polar m.i. and stress. closed coil helical springs: stress, deflection, and stiffness. stresses in thin cylindrical shells subjected to internal pressure. text / reference: t/r booktitle/author strength of materials by dr. r.k. bansal, lakshmi publishers t1 strength of materials by r.s. khurmi, s.chand& company ltd. t2 strength of materials by s.s. bhavikatti, vikas publishing house r1 r2 strength of materials by ramamrutham, dhanpat rai & sons r3 strength of materials by t.d. gunneswara rao, cambridge university press should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 4 Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press  
 ഫോറം ഫോറം definition/meaning ഫോറം ഫോറം ഫോറം,

steps, application ഏകദേശ കണക്കുകൾ ഉപയോഗിച്ച്. Technical terms English-ൽ ഉപയോഗിക്കുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

solve related problems.

|                                                                     |                  |
|---------------------------------------------------------------------|------------------|
| Learn about shafts, calculate the polar moment                      |                  |
| M4.01                                                               | 5                |
| Understanding                                                       |                  |
| of inertia (M.I.).                                                  |                  |
| Understand how springs work and calculate their                     |                  |
| M4.02                                                               | 5                |
| Understanding                                                       |                  |
| deflection and stiffness.                                           |                  |
| Understand stresses in thin cylindrical shells and                  |                  |
| M4.03                                                               | 4                |
| Understanding                                                       |                  |
| calculate their thickness.                                          |                  |
| SeriesTest-II                                                       | 1                |
| Contents:                                                           |                  |
| Torsion in shafts: calculation of polar M.I. and stress.            |                  |
| Closed coil helical springs: stress, deflection, and stiffness.     |                  |
| Stresses in thin cylindrical shells subjected to internal pressure. |                  |
| Text / Reference:                                                   |                  |
| T/R                                                                 | BookTitle/Author |
| Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers        |                  |
| T1                                                                  |                  |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

## Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

## Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

### RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

## OFFICIAL SELF-LEARNING

### Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

### Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

##### Define or explain 2 Understanding and strain.. (3 marks)

**Answer:** Begin with the standard definition or identity of 2 *Understanding and strain..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understand stress-strain diagrams for common 2 Understanding materials. Calculate stress and strain for uniform and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding composite sections. Understand thermal stresses in materials... (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding composite sections. Understand thermal stresses in materials...* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding Solve problems on stress and strain.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding Solve problems on stress and strain..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

application application application application.

## Module 2

**Define or explain Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** definition, principle/steps, application application application.

**Define or explain 2 Understanding definitions Calculate shear force and bending moments for. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding definitions Calculate shear force and bending moments for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** definition, principle/steps, application application application.

**Define or explain 10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *10 Applying various beam cases. Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams..* State the governing principle, list the key components or stages, and close



with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## Module 3

**Define or explain 2 Understanding Understand the bending equation and apply it to. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding Understand the bending equation and apply it to*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 3 Understanding simple problems.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding simple problems..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Calculate bending stress, modulus of section, and 6 Applying moment of resistance. Understand beam deflection and apply standard*. State the governing principle, list the key components or stages,

and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 2 Understanding formulae. Learn about columns and struts, and calculate. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding formulae. Learn about columns and struts, and calculate*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## Module 4

**Define or explain 5 Understanding of inertia (M.I.). Understand how springs work and calculate their. (3 marks)**

**Answer:** Begin with the standard definition or identity of *5 Understanding of inertia (M.I.). Understand how springs work and calculate their*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and. (3 marks)**

**Answer:** Begin with the standard definition or identity of *5 Understanding deflection and stiffness. Understand stresses in thin cylindrical shells and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 *Understanding calculate their thickness. SeriesTest-II 1 Torsion in shafts: calculation of polar M.I. and stress. Closed coil helical springs: stress, deflection, and stiffness. Stresses in thin cylindrical shells subjected to internal pressure. Text / Reference: T/R BookTitle/Author Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers T1 Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd. T2 Strength of Materials by S.S. Bhavikatti, Vikas Publishing House R1 R2 Strength of Materials by Ramamrutham, Dhanpat Rai & Sons R3 Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **2 Understanding and strain..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **Understand different types of beams and loads. 3 Understanding Learn shear force and bending moment.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **2 Understanding Understand the bending equation and apply it to.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **5 Understanding of inertia (M.I.). Understand how springs work and calculate their.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

# Exam checklist

## Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

## Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

## REFERENCE VOCABULARY

# Glossary

### Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## **Web resources listed in the official PDF**

No separate web-resource heading was detected in the extracted text.

### **IN-PAGE SEARCH**

## **Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 4461. Verify institutional updates against the current official curriculum.